

Computer Workshop (CMP 122)

Detailed Study Guide | Syllabus Units 1 through 8

Chapter 1: Introduction to Desktop/Workstation Assembly

1.1 Core Hardware Components

- **Central Processing Unit (CPU):** Serves as the primary computational engine, executing instructions contained within software programs through the fetch-decode-execute cycle. Key parameters include core count, thread count (via multithreading/hyper-threading), base and boost clock frequencies (measured in GHz), cache tiers (L1, L2, L3), and Thermal Design Power (TDP). Connects to the motherboard via specific socket interfaces—Land Grid Array (LGA, where pins are on the motherboard socket) or Pin Grid Array (PGA, where pins are on the CPU bottom).
- **Motherboard:** The central printed circuit board (PCB) that routes electrical power, data signals, and communication protocols between all hardware components. Form factors govern physical dimensions and mounting hole positions: ATX (standard full-size), Micro-ATX (compact with fewer expansion slots), and Mini-ITX (ultra-small form factor). Features the CPU socket, DIMM slots for memory, expansion slots (PCI Express), internal storage headers, and the chipset (which manages high-speed and low-speed peripheral communications).
- **System Memory (RAM):** Volatile high-speed temporary storage used by the CPU to hold working data and actively executing program instructions. Measured in capacity (Gigabytes) and transfer rate (MegaTransfers per second, MT/s). Modern platforms utilize DDR4 or DDR5 generations, requiring dual-channel configuration (installing matched pairs in specific slots) to maximize bandwidth.
- **Storage Subsystems:**
 - *Hard Disk Drives (HDD):* Magnetic storage devices using mechanical platters and read/write heads. Offers low cost per gigabyte but high latency and mechanical vulnerability.
 - *Solid State Drives (SSD):* Non-volatile flash memory storage. Available as SATA SSDs (capped at ~550 MB/s due to bus limits) and NVMe M.2 SSDs (utilizing PCI Express lanes directly to achieve speeds exceeding 7000 MB/s).
- **Input/Output (I/O) Devices:** Input hardware includes keyboards, mice, scanners, and digitizers that convert user interaction into digital signals via USB interfaces or legacy PS/2 ports. Output hardware includes display monitors (connecting via HDMI, DisplayPort, or USB-C), audio interfaces/speakers, and printers.

1.2 Power Systems, Cabling, and Ports

- **Switched-Mode Power Supply (SMPS / PSU):** Converts high-voltage alternating current (AC) from wall outlets into steady, low-voltage direct current (DC) rails (+3.3V, +5V, +12V) required by internal microelectronics. Categorized by total continuous wattage output and efficiency certifications (80 Plus ratings ranging from Standard to Titanium).
- **Internal Cabling Architecture:**
 - *24-Pin ATX Main Power:* Delivers primary board power across multiple voltages.
 - *4+4 Pin EPS (+12V CPU):* Dedicated power rail feed located near the CPU socket to deliver clean current to the Voltage Regulator Modules (VRMs).
 - *6+2 Pin PCIe:* Supplemental power supply cabling for dedicated Graphics Processing Units (GPUs) or high-draw expansion cards.
 - *SATA Power & Data:* 15-pin power cables from the PSU paired with 7-pin data cables connecting storage drives to motherboard headers.

- *Front Panel Connectors:* Individual small-gauge wires linking the chassis power button, reset switch, activity LEDs, and front audio/USB ports to the motherboard system pin header.
- **External I/O Ports:** Display interfaces (HDMI, DisplayPort), USB standards (USB 2.0, USB 3.2 Gen 1/2, USB4/Thunderbolt), and Networking & Audio (RJ-45 Ethernet ports, 3.5mm analog jacks, optical TOSLINK).

Chapter 2: Assembling Desktop Computer

2.1 Assembly Sequence and Safety Precautions

- **Electrostatic Discharge (ESD) Mitigation:** Static electricity accumulated on a technician's body can discharge into sensitive microconductors, causing immediate component failure or latent silicon degradation. Technicians must use an ESD grounding wrist strap attached to an unpainted metal chassis surface or work on an anti-static mat plugged into a grounded outlet circuit.
- **Workplace & Personal Safety:** Always disconnect the main power cord prior to touching internal components. Work on a flat, non-conductive, well-lit surface. Remove metal rings, watches, or loose clothing that could snag or short components.
- **Optimal Assembly Sequence:**
 1. Prepare the motherboard outside the case on top of its non-static cardboard box.
 2. Install CPU, RAM, and M.2 NVMe drives onto the unmounted motherboard.
 3. Install motherboard I/O shield (if not integrated) into the chassis rear cut-out.
 4. Insert brass motherboard standoffs in the case matching the board's form factor holes.
 5. Mount the power supply unit (PSU) and route preliminary main cabling.
 6. Lower the motherboard onto the standoffs and secure with appropriate screws.
 7. Install secondary storage drives (HDDs/SATA SSDs) and optical drives in case bays.
 8. Connect all internal power cables and front panel headers.
 9. Install expansion cards (Dedicated GPU) into the primary PCIe slot.
 10. Perform cable management for clean airflow before closing chassis panels.

2.2 Component Installation Mechanics

- **Processor (CPU) Mounting:** Release the socket load lever on the motherboard. Align the physical key notches and gold triangle indicator on the CPU corner with the matching alignment mark on the socket. Lower the CPU straight down into the socket without applying lateral force or dropping it. Close the load plate and lock down the retention lever. Apply a thermal compound layer (pea-sized dot or thin spread) onto the Integrated Heat Spreader (IHS) before clamping down the CPU cooler/heatsink assembly.
- **Memory Slotting:** Push open the retention clips on the motherboard's DIMM slots. Align the off-center notch on the bottom of the RAM module with the key in the slot. Apply firm, even downward pressure on both edges of the RAM stick simultaneously until the retention clips automatically snap shut with a click.
- **Storage & Optical Drive Mounting:**
 - *M.2 Drives:* Insert at a 30-degree angle into the M.2 slot, press down flat against the standoff, and secure with the small M.2 retention screw or toolless latch.
 - *2.5"/3.5" Drives:* Slide drives into their respective internal drive bays; fasten using UNC 6-32 screws (for 3.5" HDDs) or M3 screws (for 2.5" SSDs), or slide into toolless drive caddies.
- **Power Supply Unit Mounting:** Orient the PSU fan toward the case ventilation grill (usually pointing downward in bottom-mount cases) to ensure an isolated intake loop. Secure the PSU to the back panel of the chassis using four hex-head UNC 6-32 screws.

Chapter 3: Assembling Laptop

3.1 Disassembly, Assembly, and Precautions

- **Preparatory Protocol:** Shut down the operating system completely (avoid sleep or hibernate modes). Disconnect the external AC power adapter and all peripheral devices. If the model features a removable external battery, unlatch and slide it out prior to chassis opening.
- **Chassis Disassembly Procedure:** Map and group screws as they are removed; laptop casing screws vary widely in thread pitch, length, and diameter. Installing a long screw into a short hole can pierce internal circuit traces or crack the top palmrest surface. Use a thin plastic spudger or prying tool to release internal plastic retaining clips along the seam where the bottom cover meets the top palmrest assembly.
Critical First Action: Once inside, immediately disconnect the internal battery connector from the motherboard to kill standby power across the board before touching any other component.
- **Assembly Procedure:** Reverse the disassembly sequence. Ensure all internal ribbon cables (keyboard, trackpad, display EDP cable) are fully seated and their ZIF (Zero Insertion Force) locking latches are locked before snapping case plastics back together.

3.2 Module-Level Component Handling

- **System Board (Motherboard):** Highly integrated PCB containing soldered system components. Handle only by the edges to prevent flexing, which can crack BGA (Ball Grid Array) solder joints under major microchips.
- **SO-DIMM Memory Installation:** Small Outline DIMMs operate at a lower physical profile than desktop memory. Align the notch, insert the module into the socket at approximately a 30-degree angle, then push the top edge down toward the motherboard until side spring clips click outward and lock the board flat.
- **Laptop Storage Modules:** Disconnect internal drive ribbons or remove M.2 drives from their compact horizontal slots. Ensure non-conductive insulation shields or thermal pads are re-applied over SSD surfaces to prevent electrical contacts with metal chassis parts and dissipate heat.
- **Cooling System Maintenance:** Laptop heat dissipation relies on copper heat pipes linked to a compact blower fan. When reattaching a heatsink over a laptop CPU/GPU, clean off old thermal paste with high-purity Isopropyl Alcohol (IPA), reapply fresh thermal interface material, and tighten retention screws strictly according to the numbered sequence printed on the heatsink arms to maintain even surface contact.

Chapter 4: Laptop/Desktop OS Installation

4.1 System Firmware and Boot Configuration

- **BIOS vs. UEFI Architecture:**
 - *Legacy BIOS (Basic Input/Output System):* Uses 16-bit real mode execution, limited to initializing drives using the MBR partition scheme (maximum 2TB disk size, 4 primary partitions).
 - *UEFI (Unified Extensible Firmware Interface):* Modern 32-bit or 64-bit firmware operating environment. Supports large drives via GPT, offers graphical user interfaces, enables fast boot times, and enforces security features like **Secure Boot** (which prevents unauthorized drivers or bootloaders from executing at startup).
- **Configuring Boot Environment:** Enter firmware setup during early POST by pressing designated keys (F2, Del, F12, Esc). Set the **Boot Sequence Priority** to read from the target Installation Media (e.g., USB Flash Drive) ahead of internal hard drives. Ensure controller operational modes are set correctly (e.g., AHCI mode for SATA drives, or NVMe/RAID mode as required by platform specs).

4.2 Disk Partitioning and Operating System Deployment

- **Partition Schemes:**

- *MBR (Master Boot Record)*: Legacy standard storing boot code in the first sector of the drive.
- *GPT (GUID Partition Table)*: Modern standard required by UEFI systems. Supports up to 128 primary partitions in Windows and storage sizes spanning zettabytes, featuring redundant backup partition tables at the end of the disk.

- **Step-by-Step OS Installation Workflow:**

1. Boot system from created bootable USB drive containing OS deployment files (created via tools like Rufus or Media Creation Tool).
2. Select language, time/currency formats, and keyboard layout preferences.
3. Select *Custom/Advanced Installation* type to manage drive partitions directly.
4. In the partition manager, delete pre-existing partitions on target drives if performing a clean install, or allocate raw unallocated space.
5. Select target space and allow installer to auto-create system reserved partitions (EFI System Partition, MSR Partition, and Primary OS Recovery Partition).
6. Wait for system file extraction, feature installation, and automatic system reboot sequence.

4.3 User Account Management and Device Setup

- **User Account Provisioning:** Create primary administrative account (local user or domain connected account) with robust authentication credentials. Establish standard user accounts for daily tasks to adhere to the Principle of Least Privilege (PoLP) and prevent unauthorized software modification.
- **Device Driver Installation:** Open Device Manager to identify missing hardware drivers (indicated by yellow exclamation marks). Install dedicated Chipset, Graphics, LAN/Wi-Fi, and Audio drivers sourced from OEM manufacturer support pages.
- **Printer & Scanner Integration:** Connect peripheral hardware via local USB or network IP address. Install vendor print driver packages or configure standard Windows/Linux printing systems (such as CUPS on Linux platforms). Configure print queues, preferences, and scanner imaging utilities.

Chapter 5: Networking and Internet Setup

5.1 Network Topologies and Physical Media

- **Network Topologies:**

- *Star Topology*: Most ubiquitous modern layout. All endpoint nodes connect individually to a central distribution device (Network Switch). A single cable failure affects only one device; central switch failure drops the entire segment.
- *Bus Topology*: Legacy structure where nodes connect to a common linear coaxial backbone with terminators at both ends.
- *Ring Topology*: Data travels sequentially from node to node in a closed loop.
- *Mesh Topology*: Every node maintains dedicated links to other nodes (Full Mesh) or multiple nodes (Partial Mesh), delivering maximum redundancy and fault tolerance at higher cabling costs.

- **Physical Transmission Media:**

- *UTP (Unshielded Twisted Pair)*: Standard networking cable utilizing twisted wire pairs to cancel out electromagnetic interference (EMI) and crosstalk. (Cat5e rated up to 1 Gbps, Cat6 rated up to 10 Gbps for short distances, Cat6a rated for 10 Gbps up to 100 meters).

- *STP (Shielded Twisted Pair)*: Includes conductive foil wrapping around pairs/cables to protect data integrity in high-interference industrial environments.
- *Fiber Optic Cables*: Transmits data via light pulses over glass strands, immune to EMI, supporting extreme distance runs (Single-mode for long distance; Multi-mode for short backbones).

5.2 Cabling Termination and Wireless Setup

T568B Wiring Pinout Standard (RJ-45 8P8C):

Pin Number	Wire Color Standard (T568B)
Pin 1	White / Orange
Pin 2	Orange
Pin 3	White / Green
Pin 4	Blue
Pin 5	White / Blue
Pin 6	Green
Pin 7	White / Brown
Pin 8	Brown

- **Cable Configurations:** *Straight-Through Cable* uses the same pinout (T568B to T568B) on both ends to connect dissimilar devices (PC to Switch). *Crossover Cable* uses T568A on one end and T568B on the other to connect similar devices directly (PC to PC).
- **Wireless Access Point (WAP) Configuration:** Connect to AP management interface via default IP address over a web browser. Change default administrator credentials immediately. Set Service Set Identifier (SSID)—the broadcasted name of the wireless network. Configure wireless security encryption standards (prefer **WPA3-Personal** or **WPA2-AES**; avoid obsolete WEP or WPA-TKIP protocols). Select non-overlapping radio frequency channels (Channels 1, 6, or 11 in 2.4 GHz spectrum). Configure DHCP scope or assign a fixed static IP address to the AP.

Chapter 6: Software Installation

6.1 Application and Utility Deployment

- **Application Installation Formats:**
 - *Windows Systems*: Executable setup files (.exe) or Windows Installer packages (.msi). Executables contain compressed archives and installation scripts; .msi files leverage the native Windows Installer service for enterprise deployment.
 - *Linux Systems*: Software distributed as pre-compiled binary packages managed by distribution package managers (.deb for Debian/Ubuntu using apt; .rpm for RHEL/Fedora using dnf or yum).
- **Essential Utility Installations:** File archivers (7-Zip, WinRAR for managing .zip, .7z, .tar, .gz), system maintenance tools, disk partition managers, and PDF readers.

6.2 Software Patch Management and Security Maintenance

- **Firmware and Driver Updates:** Download official vendor flash binaries onto a clean FAT32 USB drive. Access built-in flash utilities in UEFI settings (EZ Flash, Q-Flash) to rewrite system ROM board firmware to fix memory instability and security flaws.
- **Antivirus Deployment and Malware Scanning Workflow:**
 1. *Engine Initialization:* Enable real-time background file monitoring engines.
 2. *Definition Update:* Configure automated daily sync tasks to pull updated threat signature databases over the web.
 3. *Scanning Routines:* Run quick scans on high-risk directories, full-system disk sweeps, and contextual scans for external USB devices upon insertion.
 4. *Remediation:* Automatically move detected malicious payloads to an isolated system **Quarantine** vault, block outbound IP traffic, and log infection details.

Chapter 7: Server Installation and Backup

7.1 Server Fundamentals and Architecture

- **Enterprise Server Systems:** Built for high availability, continuous multi-user throughput, and structural fault tolerance. Features multi-socket server CPU architectures (Intel Xeon, AMD EPYC), **ECC (Error-Correcting Code) Memory** capable of detecting and correcting single-bit memory corruption errors in real time, hot-swappable hard drive arrays, redundant power supplies wired to separate circuits, and multi-gigabit network interfaces. Operating systems include headless Linux Server (Ubuntu Server, RHEL) or Windows Server environments.

7.2 Network Setup and Resource Sharing

- **Static IP Configuration:** Servers must not rely on dynamic IP lease assignments. Manual configuration requires establishing: Fixed IPv4/IPv6 Address, Subnet Mask, Default Gateway, and Primary/Secondary DNS server addresses.
- **File and Folder Sharing:** Network protocols include **SMB/CIFS** (Windows standard) and **NFS** (UNIX/Linux standard). Requires configuring two permission layers: *Share Permissions* (network layer) and *NTFS/POSIX File System Permissions* (local storage layer). Access defaults to the most restrictive combination.
- **Network Printer Sharing:** Configure central print server instances, install host spooler drivers, share device onto local network, and assign Active Directory access control permissions.

7.3 Enterprise Server Roles

- **Domain Controller / Directory Server:** Runs directory services like Active Directory Domain Services (AD DS) or OpenLDAP. Centralizes identity management, user authentication, and group security policies across an enterprise domain network.
- **Data and File Server (RAID Configurations):**
 - *RAID 0:* Striping without fault tolerance (high speed, zero redundancy).
 - *RAID 1:* Disk mirroring (data duplicated across 2 drives).
 - *RAID 5:* Block-level striping with distributed parity (requires min. 3 drives; survives 1 drive failure).
 - *RAID 10:* Stripe of mirrored arrays (combines RAID 0 speed + RAID 1 redundancy; requires min. 4 drives).
- **FTP / SFTP Server:** Handles file transfers. FTP operates unencrypted over Port 21; **SFTP (SSH File Transfer Protocol)** encrypts credentials and data over Port 22.

- **Email Server:** Processes incoming/outgoing electronic mail using SMTP (Port 25/587 for sending), IMAP (Port 993 SSL for sync retrieval), and POP3 (Port 995 SSL for single-client downloading).

Chapter 8: Hardware and Software Troubleshooting

8.1 Diagnostics and Preventive Maintenance

- **Preventive Maintenance Protocol:** Periodically remove internal dust buildup using dry compressed air or specialized ESD-safe electric blowers to prevent thermal throttling. Clean and replace dried-out CPU/GPU thermal paste every few years. Maintain environment temperature/humidity limits.
- **Power-On Self-Test (POST) Diagnostics:** Motherboard firmware executes a POST sequence to check critical hardware (CPU, RAM, GPU, Storage). Outlines system status via legacy audio *Beep Codes* or modern onboard diagnostic *LED Indicators* (CPU, DRAM, VGA, BOOT).

8.2 Passive Component Failure and Hardware Repair

- **Identifying Passive Component Failures:** Electrolytic capacitors are susceptible to age/heat degradation; failed capacitors exhibit bulging tops, brown electrolyte leaks, or ruptured cross-vents. Blown SMD fuses or charred resistors indicate prior short circuits.
- **Board-Level Component Replacement Awareness:** Requires micro-soldering equipment, liquid rosin flux, braided solder wick, temperature-controlled iron, and hot-air rework station. Technicians must desolder broken components without lifting tiny copper PCB pads and replace them with identical spec passives.

8.3 Diagnostic Commands & Toolsets

Essential Command Line Diagnostic Utilities

Windows CLI Tools:

```
ping [IP_Address / Hostname]      # Tests network layer connectivity & latency
ipconfig /all                     # Displays complete network interfaces, MAC, DNS,
DHCP specs
sfc /scannow                      # System File Checker - scans and repairs
corrupted Windows files
chkdsk C: /f /r                  # Disk repair - (/f) fixes logical errors, (/r)
recovers bad sectors
tracert [Destination_IP]         # Traces network path hops & identifies latency
bottlenecks
```

Linux CLI Tools:

```
ip a / ifconfig                  # Displays active network adapters and IP
configurations
ping [IP_Address]                # ICMP packet echo test (use Ctrl+C to stop)
top / htop                       # Real-time resource monitor (CPU, Memory, PID
processes)
dmesg                            # Prints kernel ring buffer logs for hardware/
driver diagnostics
```